

Experiment 6: Depth First Search (DFS)

Aim

To implement Depth First Search (DFS) using Python.

Objective

To traverse all nodes of a graph using DFS.

Algorithm

1. Create a graph.
2. Start from the source node.
3. Visit the node.
4. Recursively visit all unvisited neighbours.
5. Repeat until all nodes are visited.

Flowchart

Start

|

Create Graph

|

Visit Node

|

Unvisited Neighbour?

/ \

Yes No

|

|

Visit Backtrack

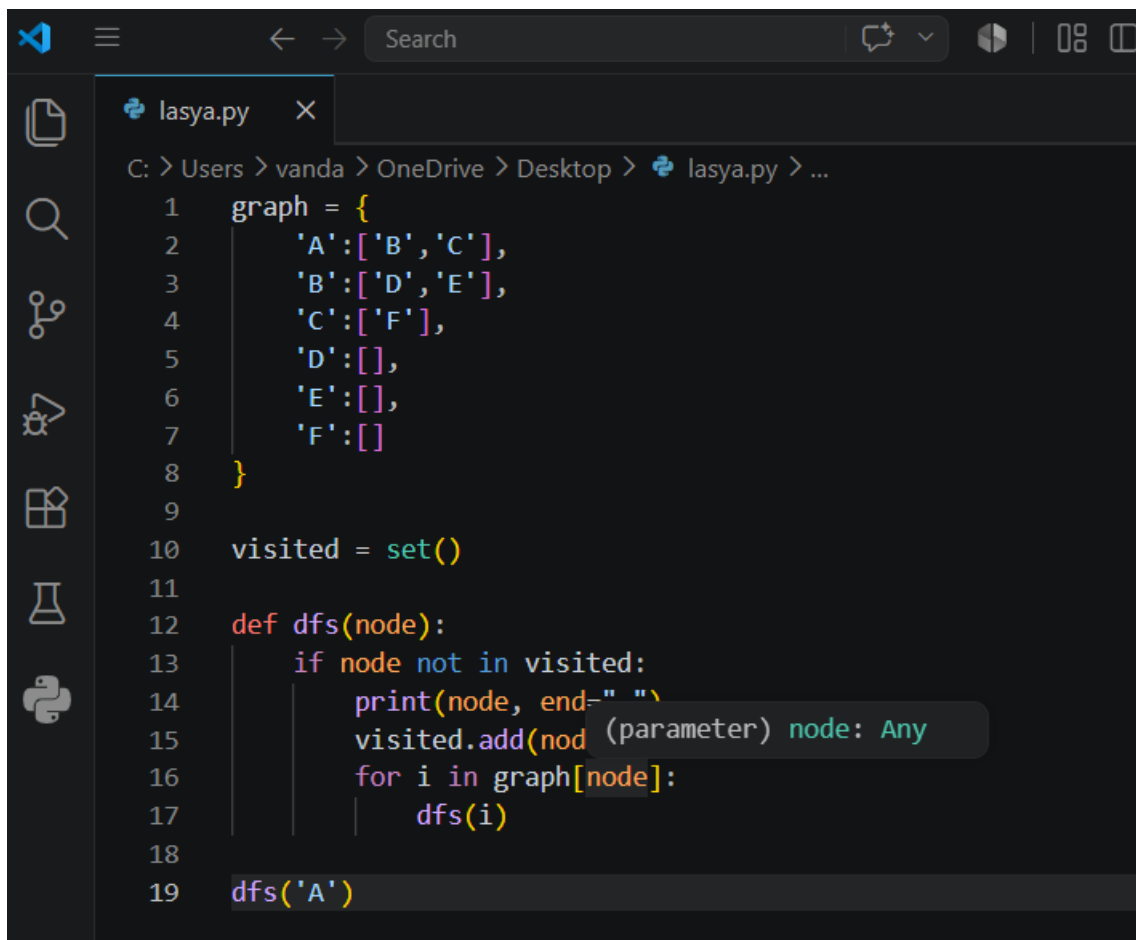
\ /

Repeat

|

Stop

Code



```
1 graph = {
2     'A': ['B', 'C'],
3     'B': ['D', 'E'],
4     'C': ['F'],
5     'D': [],
6     'E': [],
7     'F': []
8 }
9
10 visited = set()
11
12 def dfs(node):
13     if node not in visited:
14         print(node, end=" ")
15         visited.add(node)
16         for i in graph[node]:
17             dfs(i)
18
19 dfs('A')
```

Output:

```
PS C:\Users\vanda\onedrive\desktop> python lasya.py
A B D E C F
PS C:\Users\vanda\onedrive\desktop> 
```

Result

The graph was traversed successfully using DFS.

Conclusion

DFS visits nodes by exploring each branch completely before backtracking.